Endocrine System

The endocrine system is composed principally of glands that have lost connection with the epithelium and, therefore, have no ducts. Their secretions (hormones) are released directly into the blood. Their rich blood supply not only serves their metabolic needs but also functions to transport their secretory products. Endocrine tissues include the pituitary, thyroid, parathyroid, adrenal and pineal glands as well as the pancreatic islets (of Langerhans) and scattered cells of the diffuse neuroendocrine system (DNES).

- Endocrine glands have a simple histological structure: they consist of either cords or clumps of cells separated by capillaries or sinusoids that are supported by delicate connective tissue. Each gland secretes one or more types of **hormones**.
- Secretory granules within the cells contain hormones or their precursors and most require special staining methods to be seen. Secretory products may be stored extracellularly as in a thyroid follicle or released immediately after formation as in the adrenal cortex.

Laboratory Experience

- The purpose of this laboratory exercise is to identify the different types of cells and tissues in the endocrine system and to describe the hormones produced and their function.

Pituitary: Slides # 4 – 11 (H&E & trichrome) + diagrams

- The **anterior pituitary** (pars distalis or adenohypophysis) is derived from oral ectoderm which migrates dorsally as **Rathke's pouch** and engulfs the **posterior pituitary** (pars nervosa or neurohypophysis) which is derived from the midbrain. The **pars intermedia** is composed of basophilic cells (Secrete MSH), located between the anterior and posterior pituitary.

- The parenchyma of the **anterior pituitary** contains cords of two types of cells; **chromophils** and **chromophobes** supported by a reticular network and fenestrated capillaries that lie between the cords. **Chromophils** are classified as **acidophils** or **basophils** based on their staining reactions.

- **Acidophils** contain small granules which stain eosinophilic. They can be mammotrophs (producing and secreting **prolactin (PRL)**) or somatotrophs (producing and secreting **growth hormone (GH)**).

- **Basophils** contain granules that stain basophilic. They can be gonadotrophs (producing **luteinizing hormone (LH)** or **follicle stimulating hormone (FSH)**), adrenocorticotrophs (secreting **adrenocorticotropic hormone (ACTH)**) or thyrotrophs (secreting **thyrotropin stimulating hormone (TSH)**).

- The **posterior pituitary** contains blood vessels, unmyelinated nerve fibers and neuroglial **pituicytes**. Neurons in the supraoptic and paraventricular nuclei of the hypothalamus produce **oxytocin** and **vasopressin** (**anti-diuretic hormone (ADH)**). These hormones are transported through unmyelinated nerve fibers to nerve terminals within the posterior pituitary where they are stored prior to release. Large accumulations of the stored hormones may be visible with light microscopy as **Herring bodies**.

Identify and check-off each of the following:

( ) Anterior pituitary **( ) Chromophobe** **( ) Acidophil** **( ) Basophil**

( ) Posterior pituitary **( ) Herring body** **( ) Pituicyte** **( ) Pars intermedia**

Pancreas: Slides # 12-13 (H&E + Grimelius silver stain)

- Review the exocrine acini and ducts of the pancreas. Identify the pale stained **pancreatic islets** (of Langerhans) interspersed between the exocrine tissue.

Special staining and ultrastructural methods are required to distinguish between **alpha**, **beta** and **delta** cells that secrete, respectively, **glucagon**, **insulin** and **somatostatin**. Alpha cells tend to be found at the periphery of the islet with beta cells at the center.

Identify and check-off each of the following:

( ) Pancreatic islets (of Langerhans) **( ) Exocrine acini (serous)**
( ) Exocrine ducts

Pineal Gland: Slides # 14-15 (H&E)

- The pineal body (**epiphysis cerebri**), is covered by a capsule of pia mater which extends into the organ dividing it into incomplete lobules. The irregular lobules are composed of cells called **pinealocytes** and supporting neuroglial cells (astrocytes).

- With age, extracellular **pineal concretions** (acervuli, corpora arenacea or brain sand) appear; these are calcified bodies that are visible with light microscopy and on radiographs. Pinealocytes produce **melatonin** which influences circadian rhythms; its production is affected by light / dark cycles.

Identify and check-off each of the following:

( ) Pinealocytes **( ) Pineal concretions** **( ) pia mater**

Adrenal Gland: Slides # 16-20 (H&E) + diagrams

- The adrenal glands have a cortex and medulla. The cortex is divided into three layers: a thin outer, **zona glomerulosa (10-15%)**; a thick middle **zona fasciculate (75%)**; and an inner **zona reticularis (5%)**. Numerous fenestrated capillaries are present. The zona fasciculata has numerous light staining cells (spongiocytes) that are steroid producing and contain large quantities of glycogen and smooth endoplasmic reticulum.

- **Mineralocorticoids** (that control electrolyte and water balance) are produced in the zona glomerulosa. **Glucocorticoids** are synthesized in the zona fasciculata and in the zona reticularis (that act on carbohydrate metabolism and suppress immune responses). **Androgens** or their precursors are produced in the zona reticularis. These hormones are regulated by ACTH release from the anterior pituitary.

- The **medulla** is derived from the neural crest and is highly vascular with fenestrated capillaries. Its cells (**chromaffin cells**), which are modified post-ganglionic sympathetic neurons, contain granules which secrete epinephrine and/or norepinephrine. These cells stain brown with chromium salts (potassium dichromate). They also contain few ganglion cells that send their axons to the cortex (blood vessels) and neighboring structures.

Identify and check-off each of the following:

( ) Capsule ( ) Cortex ( ) Zona glomerulosa ( ) Zona fasciculata ( ) Zona reticularis ( ) Medulla ( ) Chromaffin cells ( ) Blood vessels

Thyroid Gland: Slides # 22-23 (H&E) **+ diagrams**

- A capsule of connective tissue extends to divide the thyroid gland into **lobules** which contain **follicles**. Each follicle consists of a simple cuboidal epithelium (follicle cells) enclosing a cavity filled with **thyroglobulin**, the stored precursor of thyroxine and triiodothyronine.

- The two functional cell types are the cuboidal **follicular cells** (principal cells) and the pale **parafollicular** (clear cells) located at the periphery of the follicles, in the stroma.

- Principle cells produce colloid and thyroid hormone, while parafollicular cells produce and secrete calcitonin which lowers blood calcium levels (act on osteoclasts and suppress their activity). Recall that thyrotropin stimulating hormone from the anterior pituitary stimulates thyroid hormone production and release.

Identify and check-off each of the following:

( ) Follicular cells ( ) Colloid ( ) Parafollicular cells ( ) Capillaries ( ) Follicles

Parathyroid gland: Slides # 25 -28 (H&E)

- The parathyroid glands are embedded in the wall of the thyroid. They are composed of cords of epithelial cells supported by reticular fibers and a rich network of fenestrated capillaries.

- The principal or **chief cells** are small and basophilic and produce **parathyroid hormone** which raises blood calcium levels (act on osteoblasts to produce osteoclast stimulating factors). The **oxyphil cells** are larger than the chief cells, have an eosinophilic cytoplasm and an unknown function.

Identify and check-off each of the following:

( )Capsule	**( )Oxyphil cells**	**( )Blood vessels**	**( )Chief cells**

Study Questions:

Acidophils produce two hormones; _______________ and ____________.

Basophils produce four hormones; _________, ____________, ___________ and _________.

Parafollicular cells produce _____________ which functions to _______________.

The function of oxyphils is _____________.

The zona glomerulosa produces what hormone?

What cells produce epinephrine in the adrenal gland? What is the embryological origin of adrenal medulla?

Alpha cells in the pancreatic islets are found where specifically and produce what hormone? By what technique can alpha and beta cells be visualized in histological slides?

Worksheet - Fill in the Boxes

Hormone	Source	Target(s)	Action(s)
GH			
PRL			

ACTH			
TSH			
FSH			
LH			

Hormone	Source	Target(s)	Action(s)
ADH			
Melatonin			
Aldosterone			
Cortisol			
Epinephrine			

Thyroxine			
Calcitonin			
PTH			
Glucagon			
Insulin			